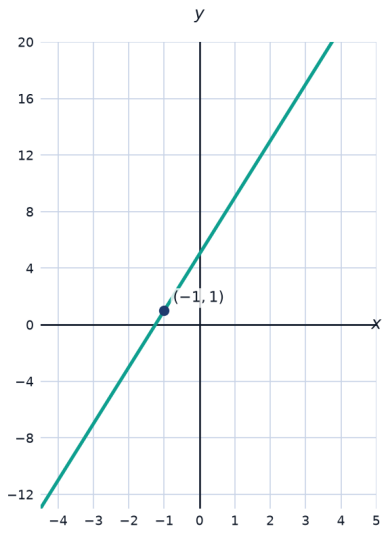
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

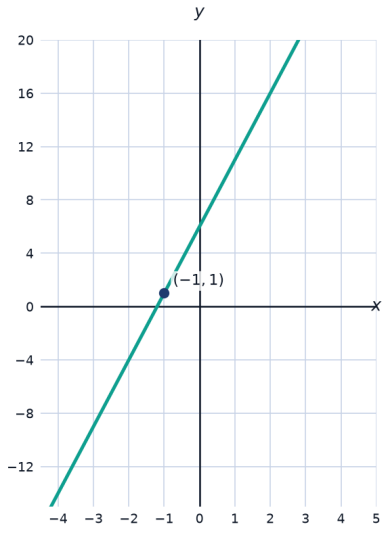
**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(5)$.




**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(2)$.



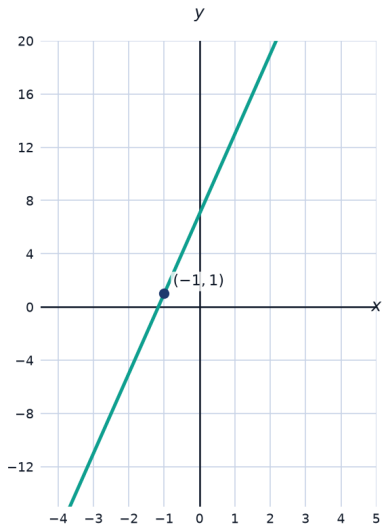

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for fractions and the number line consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

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**4** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(2)$.

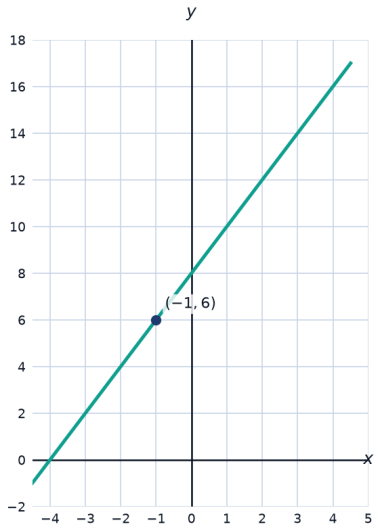



**5** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for multiplying and dividing integers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

**6** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(2)$.




**7** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

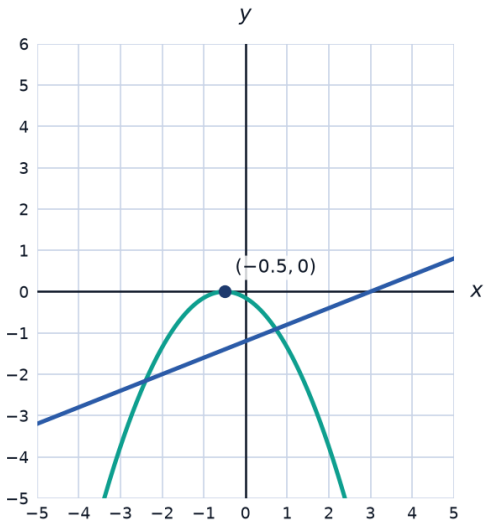
Segment $AB$ has endpoints $A(2,\ -1)$ and $B(6, 7)$. Find the equation of the perpendicular bisector of $AB$ in slope-intercept form.

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the equation of the perpendicular bisector of the line segment joining $A(-2, 5)$ and $B(4,\ -3)$. Give your answer in the form $ax + by + c = 0$, where $a, b, c$ are integers.

**10** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coordinates of the point $P$ on directed line segment from $A(-3, 2)$ to $B(7, 12)$ that partitions the segment in the ratio $2 : 3$.




**11** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the set of values of the constant $k$ for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

**12** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circular archery target has three concentric rings of radii $10\ \text{cm}$, $20\ \text{cm}$, and $30\ \text{cm}$. If an arrow hits the target completely at random, what is the exact probability that it lands in the middle ring (between radii $10\ \text{cm}$ and $20\ \text{cm}$)?

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A rhombus has diagonals of lengths $16\ \text{cm}$ and $12\ \text{cm}$. Find the perimeter and area of the rhombus.

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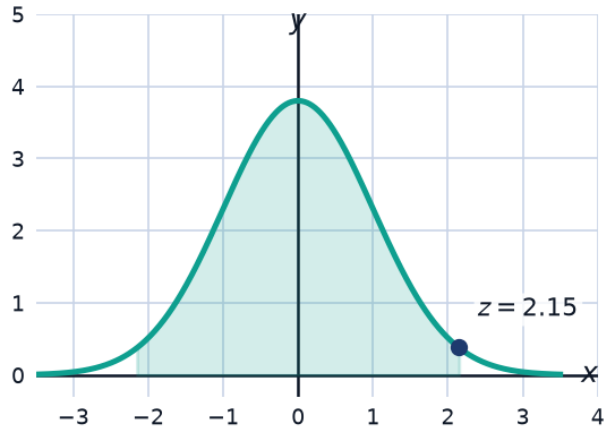
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

From an external point $P$, two tangents $PA$ and $PB$ are drawn to a circle with radius $r = 5$ cm. The distance from $P$ to the center of the circle $O$ is $13$ cm. Find the length of the tangents $PA$ and $PB$, and find the perimeter of quadrilateral $PAOB$.

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Algebra & Graphs

A block of mass $8$ kg rests in limiting equilibrium on a rough plane inclined at $30^\circ$ to the horizontal. Taking $g = 9.8$ m/s$^2$: (a) calculate the normal reaction $R$, (b) find the coefficient of friction $\mu$ between the block and the plane.




**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Coordinate Geometry

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

A train accelerates uniformly from rest to $30$ m/s in $20$ s, travels at constant speed $30$ m/s for $60$ s, and then decelerates uniformly to rest in $15$ s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

A block of mass $4$ kg is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25$ N acting at $30^\circ$ above the horizontal. Taking $g = 9.8$ m/s$^2$: (a) show that the normal reaction $R = 26.7$ N, (b) find the acceleration of the block.

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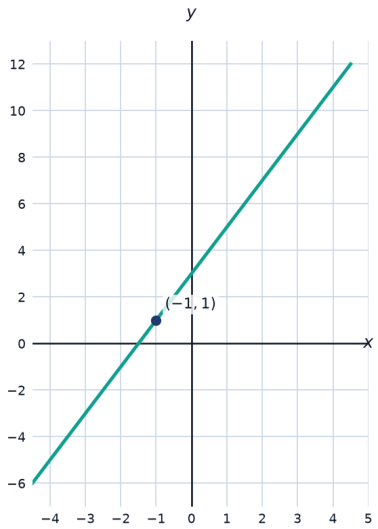
**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Transformations & Vectors

Two particles $P$ of mass $3\ \text{kg}$ and $Q$ of mass $5\ \text{kg}$ are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8\ \text{m/s}^2$: (a) find the acceleration $a$ of the system, (b) calculate the tension $T$ in the string, (c) find the force exerted by the string on the pulley.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(2)$.




**a.** 7

**b.** 9

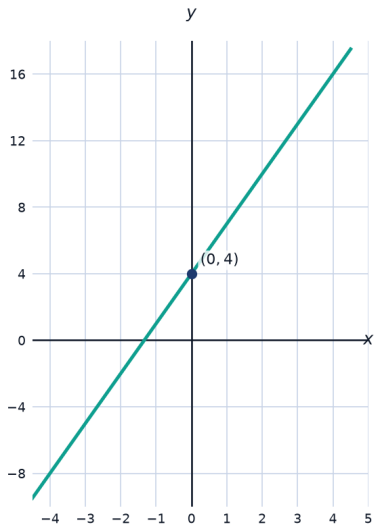
**c.** 4

**d.** 11

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**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(2)$.




**a.** 10

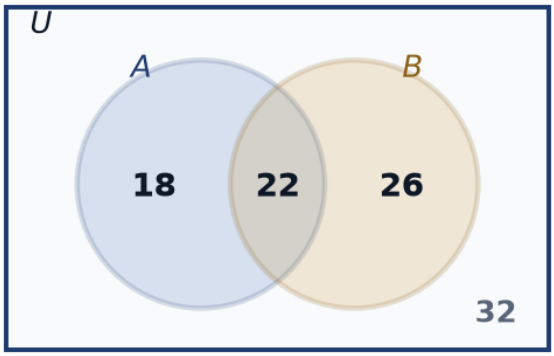
**b.** 13

**c.** 6

**d.** 14

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?




**a.** 22.5

**b.** 23.5

**c.** 24.5

**d.** 26.5

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**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting integers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** 14.5

**b.** 15.5

**c.** 16.5

**d.** 18.5

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A chord of length 24 cm is at a distance of 5 cm from the centre of a circle. What is the radius of the circle?

**a.** 13 cm

**b.** 12 cm

**c.** 14 cm

**d.** 15 cm

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A tangent TA touches a circle at A. Chord AB makes an angle of 38° with TA. Point C lies on the major arc AB. What is angle ACB?

**a.** 38°

**b.** 142°

**c.** 76°

**d.** 52°

**27** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A tangent touches a circle at point $A$. Chord $AB$ makes an angle of $48^\circ$ with the tangent. By the alternate segment theorem, what is the angle subtended by chord $AB$ in the alternate segment?

**a.** 48°

**b.** 132°

**c.** 42°

**d.** 96°

**28** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a circle with centre $O$, point $P$ lies on the circumference. What is the angle between the tangent at $P$ and the radius $OP$?

**a.** $45^\circ$

**b.** $60^\circ$

**c.** $90^\circ$

**d.** $180^\circ$

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Each interior angle of a regular polygon is $156^\circ$. How many sides does the polygon have?

**a.** 12

**b.** 15

**c.** 18

**d.** 20

**30** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Triangle ABC is inscribed in a circle such that AB is a diameter. If angle BAC = 34°, find the size of angle ABC.

**a.** 56°

**b.** 34°

**c.** 66°

**d.** 46°

**31** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Points $A, B, C, D$ lie on a circle. Chords $AC$ and $BD$ intersect at $X$. If angle $\angle ABD = 42^\circ$, what is the size of angle $\angle ACD$?

**a.** $42^\circ$

**b.** $48^\circ$

**c.** $84^\circ$

**d.** $96^\circ$

**32** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Points $A$, $B$, and $C$ lie on a circle. $AB$ is a diameter. What is the size of angle $\angle ACB$?

**a.** $45^\circ$

**b.** $60^\circ$

**c.** $90^\circ$

**d.** $120^\circ$

**33** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Points A, B, and C lie on a circle. Angle ABC = 47°. What is the reflex angle AOC subtended at centre O?

**a.** 266°

**b.** 94°

**c.** 188°

**d.** 274°

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**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In cyclic quadrilateral PQRS, angle PQR = 104°. What is the size of the opposite angle PSR?

**a.** 76°

**b.** 104°

**c.** 86°

**d.** 90°

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a regular polygon, the sum of all interior angles is 1440°. How many sides does this polygon have?

**a.** 10

**b.** 8

**c.** 12

**d.** 9

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a cyclic quadrilateral $ABCD$, angle $A = 75^\circ$. What is the size of the opposite angle $C$?

**a.** 105°

**b.** 75°

**c.** 127.5°

**d.** 90°

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The interior angle of a regular polygon is 160°. How many sides does the polygon have?

**a.** 18

**b.** 16

**c.** 20

**d.** 15

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

What is the size of each interior angle of a regular 5-sided polygon?

**a.** 108°

**b.** 72°

**c.** 123°

**d.** 98°

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**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

An exterior angle of an isosceles triangle is 110°. If the two non-adjacent opposite angles are equal, what is the measure of each of those two angles?

**a.** 55°

**b.** 70°

**c.** 35°

**d.** 60°

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Transformations & Vectors

Surface area of a sphere radius 5:

**a.** $25\pi$

**b.** $50\pi$

**c.** $75\pi$

**d.** $100\pi$

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

In a circle, chords AB and CD intersect at point P. If angle ACD = 28°, what is the size of angle ABD?

**a.** 28°

**b.** 62°

**c.** 56°

**d.** 152°

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Functions

In a regular polygon, each exterior angle is $30^\circ$. How many sides does the polygon have?

**a.** 10

**b.** 12

**c.** 14

**d.** 15

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number

Triangle DEF is an enlargement of triangle ABC with scale factor 2.5. If the area of triangle ABC is 12 cm², what is the area of triangle DEF?

**a.** 75 cm²

**b.** 30 cm²

**c.** 60 cm²

**d.** 150 cm²

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**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

Points $A, B, C$ lie on a circle with centre $O$. If angle $\angle ABC = 35^\circ$, what is the angle $\angle AOC$ subtended at the centre?

**a.** 70°

**b.** 35°

**c.** 17.5°

**d.** 145°

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Algebra & Graphs

Two similar triangles have areas in the ratio 16 : 49. What is the ratio of their corresponding perimeters?

**a.** 4 : 7

**b.** 16 : 49

**c.** 2 : 3.5

**d.** 256 : 2401

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